

Dezfouli bandit data — summary

Overview

Two-armed bandit (two-choice) behavioural data from Dezfouli et al. (2019), covering unipolar depression (MDD), bipolar disorder (BD) and healthy controls (HC). This report summarises the preprocessed dataset with group-level descriptive statistics.

Cohort

MDD 34 subjects (raw label: Depression)

BD 33 subjects (raw label: Bipolar)

HC 34 subjects (raw label: Healthy)

Total 101 subjects · 12 sequences each · 1,212 sequences · 132,251 trials.

Task design

Each subject completes 12 sequences = 6 reward-probability conditions × 2 experimental halves. Arm R1 is always the better option (reward rate ≈ 0.08 – 0.25 by condition); R2 ≈ 0.05 .

Demographics

The only subject-level variables in the released file are the diagnosis label, a mania score, a depression score and a mean press rate. No age, sex, IQ, education or medication data are included.

Provenance & license

Source: github.com/adezfouli/rnn_hypercoder (data/BD/, Apache-2.0); canonical dataset on Figshare article 8257259. Cite Dezfouli et al. (NeurIPS 2019; PLOS Comput Biol 2019).

Group descriptive statistics

Per-subject values summarised as mean $\pm$ standard deviation.

Metric	HC (n=34)	MDD (n=34)	BD (n=33)
Depression score	1.50 $\pm$ 2.06	14.32 $\pm$ 7.10	8.76 $\pm$ 6.53
Mania score	0.09 $\pm$ 0.38	2.44 $\pm$ 5.43	4.42 $\pm$ 5.84
Press rate (s)	1.65 $\pm$ 0.48	1.74 $\pm$ 0.57	1.51 $\pm$ 0.42
Reward rate	0.114 $\pm$ 0.010	0.107 $\pm$ 0.011	0.108 $\pm$ 0.009
P(better arm R1)	0.51 $\pm$ 0.06	0.51 $\pm$ 0.09	0.53 $\pm$ 0.06
Trials completed	1313 $\pm$ 333	1379 $\pm$ 383	1234 $\pm$ 292

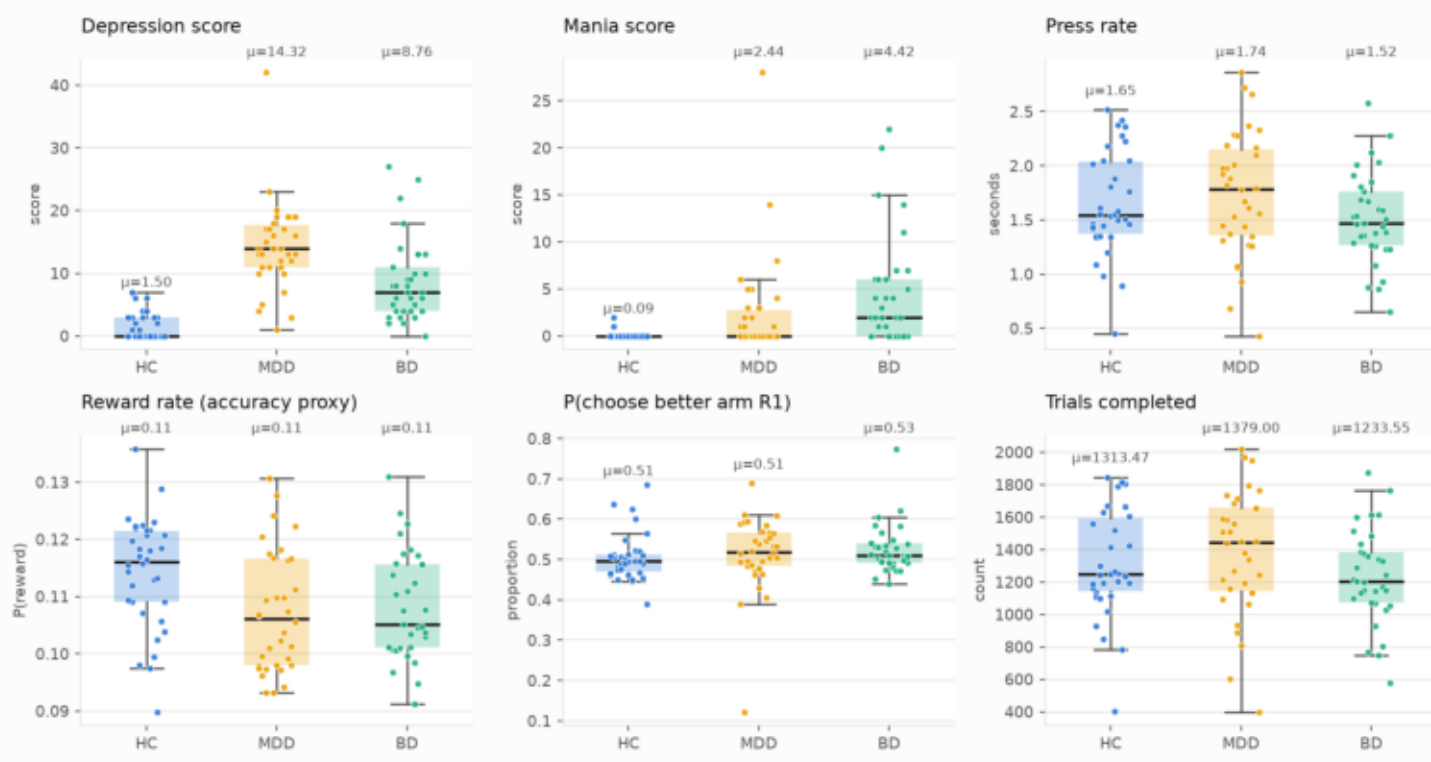
Notes:

- Clinical scores separate the groups as expected: depression highest in MDD, mania highest in BD.
- Behavioural group differences are small — the task uses low reward probabilities (R1 $\approx$ 0.08–0.25 vs R2 $\approx$ 0.05), so preference for the better arm is weak across all groups.
- Descriptive statistics only; no group-difference tests were performed.

Group descriptives (per subject)

Dezfouli bandit data — group descriptives (per subject)

● HC (healthy) ● MDD (unipolar) ● BD (bipolar)



Learning within a sequence, by group

